Chapter 20 | Written solutions

Translation and Summarization

Work through all twelve exercises. Begin with the facts a task must preserve, then align training targets, multiply path probabilities, and count reference matches. The Python investigations expose padding and stopping rules. The last four exercises connect search and source evidence to evaluation design. The written edition keeps every explanation and plot; the notebook PDF adds code and saved results. Exercises 9-12 are optional extensions.

1. Decide what must be preserved

A translation must preserve the ferry's identity as the last departure, its actual departure time of 19:10, the thirty passengers, and the gate inspection as the cause of the delay. Word order and time formatting may change with language. The factual relations must remain the same.

For a passenger asking why it was late, one possible summary is: "The last ferry departed at 19:10 after a gate inspection delayed it." This keeps the cause and actual departure time. The passenger count is incidental to that question. A different audience could need it, so the omission is a decision under the stated task, not a universal rule for summarization.

This selection is an explicit annotation of the example, not an automatic test of semantic equivalence. The notebook records the two chosen facts.

2. Align two target lengths

Target $(5, 7, 6)$ produces decoder input $(1, 5, 7, 6)$ and labels $(5, 7, 6, 2)$. Target $(8, 9)$ produces input $(1, 8, 9)$ and labels $(8, 9, 2)$ before padding. Pad the shorter rows to width four with zero:

$$Y_{\text{in}} = \begin{bmatrix} 1 & 5 & 7 & 6 \\ 1 & 8 & 9 & 0 \end{bmatrix}, \qquad Y_{\text{target}} = \begin{bmatrix} 5 & 7 & 6 & 2 \\ 8 & 9 & 2 & 0 \end{bmatrix}.$$

Read corresponding columns within a row. In the second row's third column, input 9 predicts EOS, whose ID is 2. In the fourth column, the label is padding and contributes nothing to the loss. The first row has four supervised positions and the second has three: seven in total, including both EOS labels. Teacher forcing gives each prediction the reference prefix; a causal mask blocks later decoder inputs. The loss mask handles padding separately.

3. Compare complete paths

Multiply conditional probabilities along each path, including the end marker. Choosing A and then EOS has probability $0.7(0.6) = 0.42$. Going through C adds one more factor; its EOS probability is one.

Completed path	Product	Probability
A, EOS	$0.7(0.6)$	0.42
A, C, EOS	$0.7(0.4)(1)$	0.28
B, EOS	$0.3(0.9)$	0.27
B, C, EOS	$0.3(0.1)(1)$	0.03

These four disjoint complete paths exhaust this tree and sum to one. Greedy decoding chooses A at the root and EOS after A. It happens to find the globally most probable complete path here. The chapter's different tree, with 0.60/0.40 at the root, shows that this success is not guaranteed.

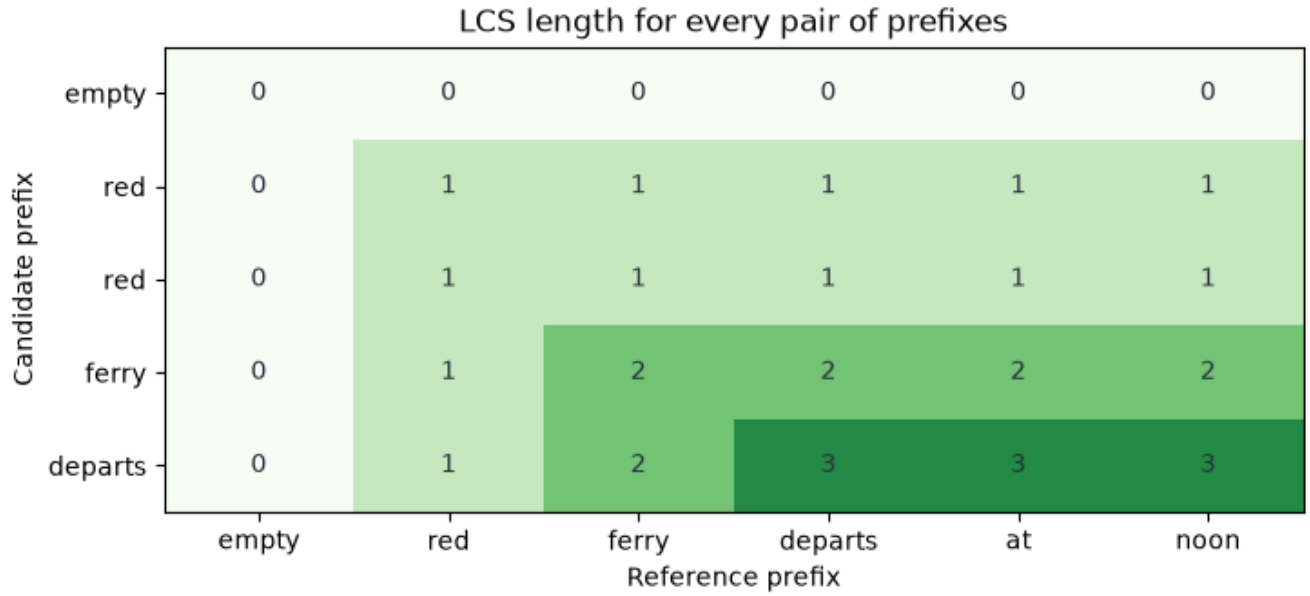
4. Count repeated and ordered matches

Candidate `red red ferry departs` has four tokens. Reference `red ferry departs at noon` has five. Clip each candidate unigram count by its reference count:

Token	Candidate count	Reference count	Clipped count
red	2	1	1
ferry	1	1	1
departs	1	1	1

The numerator is three. The denominator still counts every candidate token, including the repeated red, giving $3/4 = 0.75$. A longest common subsequence is `red ferry departs`, of length three. Either candidate red can supply the single reference red; one reference occurrence cannot be reused. The reference tokens at and noon have no candidate matches, so a longer subsequence is impossible. Matching order is required, but adjacency is not.

The following table makes the LCS calculation visible. Row i represents the first i candidate tokens; column j represents the first j reference tokens. Empty prefixes give zero. Matching last tokens add one to the diagonal predecessor; a mismatch takes the larger count from above or left. At the second red, the count stays one: there is still only one red on the reference side. The last cell gives three. The full table is for teaching; the canonical helper only retains the previous and current rows.



5. Aggregate counts before dividing

The two match counts are 3 and 4; the candidate lengths are 4 and 10. Corpus unigram precision is

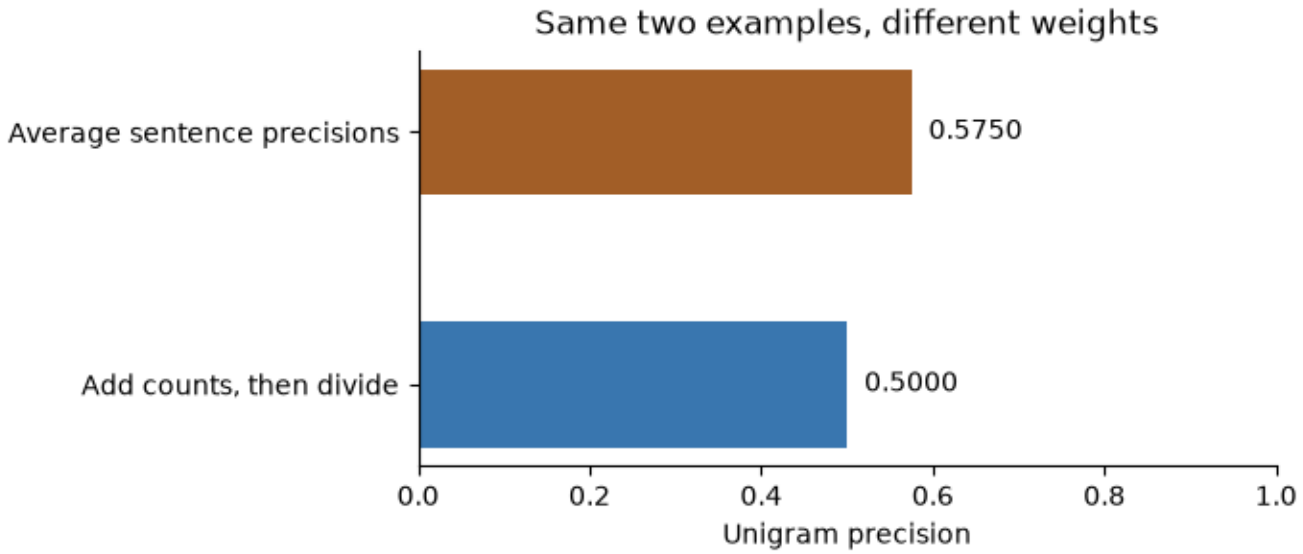
$$p_1 = \frac{3+4}{4+10} = \frac{7}{14} = 0.5.$$

The mean of the two sentence precisions is

$$\frac{1}{2}\left(\frac{3}{4} + \frac{4}{10}\right) = 0.575.$$

The corpus calculation gives equal weight to each candidate token. The sentence average gives equal weight to each example, so the four-token example has as much influence as the ten-token example. Neither quantity is complete BLEU: one also needs the specified higher n-gram orders, their geometric combination, and the brevity penalty. A zero numerator and a zero denominator are different situations. The helper returns `None` when the candidate has no n-grams of the requested order; do not turn that into a perfect match.

The bars show the two aggregations of exactly the same matches. Their difference comes from weighting, not from a new prediction or tokenizer.



6. Inspect a padded batch

Pair source `[3, 4]` with target `[5, 7, 6]` and source `[5, 6, 7, 8, 9]` with target `[8, 9]`. The encoder reads the source rows. The decoder reads target tokens shifted one position to the right of the corresponding labels. It starts with `BOS = 1` and learns to predict the end marker `EOS = 2` after the final content token.

The longest source has five tokens, so both source rows have five columns after padding. The longest target has three content tokens. Each of the two target-side arrays needs four columns, one for each content token plus one for its boundary marker. The batch has two rows in all three arrays, and `PAD = 0` fills any unused columns on the right.

A label is valid when its ID differs from `PAD`. For example, the second decoder row reads `9` in the third column and must predict `EOS` there. That position contributes to the loss, while the fourth column in the same row is padding. The loss counts all five content labels and both end markers.

7. Stop at a length cap

The first call to the probability function receives the empty tuple. This means the search has generated zero tokens. The only first choice is A, whose ID is 3. When the prefix is `(3,)`, the only next choice is the end-of-sequence token `EOS`, whose ID is 2.

A beam of width one keeps just the highest-scoring candidate at each step. One step generates A and uses up the budget. With two steps, the search can append `EOS`. The callback supplies a dictionary for each prefix that can be expanded in these runs. Once `EOS` is appended, the search considers that path complete and never asks for another token from it.

At the one-step cap, the return value is `((3,), 0.0, False)`. At the two-step cap, it is `((3, 2), 0.0, True)`. The second result includes the end token because emitting it takes a step. Both log-probabilities are zero. The one-token prefix has probability 1, and multiplying by the conditional probability 1 of `EOS` gives the same probability for the complete sequence. The flag records whether the returned tokens end in `EOS`.

8. Compare overlap measures

The candidate is *red red ferry departs* and the reference is *red ferry departs at noon*. An n-gram is a run of n adjacent tokens. A unigram is one token. For each unigram, clipping takes the smaller of its counts in the two sequences. The candidate has two occurrences of `red`, and the reference has one, so `red` contributes one match. The other matches are `ferry` and `departs`, for three in total.

A bigram is a pair of adjacent tokens in order. The candidate has `red red`, `red ferry`, and `ferry departs`. The reference has `red ferry`, `ferry departs`, `departs at`, and `at noon`. The two matching bigrams are `red ferry` and `ferry departs`.

Swapping the inputs exchanges the two counts for each n-gram. Taking their minimum gives the same result, so each clipped match total stays fixed. Precision divides by the number of n-grams in whichever sequence is now the candidate. The expected unigram precision goes from $3/4$ to $3/5$, and the expected bigram precision goes from $2/3$ to $2/4$.

ROUGE-L counts the length of a longest common subsequence. A subsequence keeps tokens in order and may skip intervening tokens. Here `red ferry departs` is common to both sequences. It starts at the second `red` in the candidate and the first `red` in the reference. It has length three, which is as long as any common subsequence can be here. Only three token occurrences can match even if we ignore their order, as the clipped unigram count showed.

Precision divides that length by the candidate length, and recall divides it by the reference length. The original precision is $3/4$ and recall is $3/5$. Swapping the inputs exchanges those values. F1 is the harmonic mean of precision and recall. Here it equals twice the common subsequence length divided by the sum of the two sequence lengths. Those quantities stay fixed after the swap, so F1 stays at $2(3)/(4+5) = 2/3$.

9. Keep a useful prefix (optional)

A starts with probability 0.65 and B starts with 0.35. With room for only one candidate, the decoder keeps A and discards B at the first step. A width of two keeps both. The score of each path is the product of its successive next-token probabilities. The implementation adds their logarithms and ranks candidates by decreasing log-probability, which gives the same order as ranking by decreasing probability.

The table in the callback gives every distribution that the four requested runs can query. An `EOS` candidate keeps its score and can still occupy a beam slot. The branch that ends in D can first be generated on step three, so the callback needs no next-token distribution after D for these step caps.

At step two, A can extend to A C with probability $0.65 \times 0.7 = 0.455$ or A EOS with probability $0.65 \times 0.3 = 0.195$. If B survived step one, it extends to B EOS with probability $0.35 \times 1 = 0.350$. Width one keeps A C. Width two also keeps B EOS, which outranks A EOS. In both runs with a two-step cap, the highest-scoring retained path is A C, so the search returns it with `finished=False`. A finished path can share the beam with a higher-scoring unfinished path.

At step three, A C extends to A C EOS with probability $0.455 \times 0.6 = 0.273$ and A C D with probability $0.455 \times 0.4 = 0.182$. Width one returns A C EOS, the higher-scoring of the two extensions. With width two, B EOS carries forward at 0.350 without another probability factor. It ranks above both extensions of A C, so the two survivors in order are B EOS and A C EOS. The returned path ends in `EOS` at both widths with the three-step cap. B EOS can win only in the run that kept B at step one.

10. Check a summary claim (optional)

The source states that the ferry was late and the north gate was closed. Replacing and with because adds a third proposition:

Candidate claim	Source assessment
Ferry was late	Supported directly
North gate was closed	Supported directly
Gate closure caused the delay	Unsupported: the cause is unstated

Unsupported does not mean proven false. It means this source does not establish the causal relation. Read the candidate against the notice before scoring it. Using lowercase word tokens and ignoring punctuation, source and candidate each have ten tokens. Nine appear in the same order; skip and on one side and because on the other. LCS length nine gives precision, recall, and F1 all equal to 0.9. A high overlap score therefore coexists with an invented causal link. The code confirms the overlap calculation; the claim labels come from reading the source.

11. Trace a missing fact (optional)

Ask: "In one sentence, name the departure gate of the replacement ferry." Use the answer-bearing sentence "The replacement ferry departs from gate 7." Other sentences discuss weather or ticket sales and contain no other gate numbers. The required fact is the relation between this ferry and gate 7, regardless of the output wording.

For a real tokenizer, choose an encoder limit such as 128 tokens. Keep the instruction at the front and use right truncation. Construct early and late versions of the same notice by swapping the answer sentence with irrelevant material. In the early version the whole answer sentence must survive; in the late version it must be wholly beyond the limit. Verify the positions after formatting and tokenization, including special tokens. Avoid a partially truncated answer for this first comparison.

Inspect the actual input immediately before the encoder call. Record the formatted text, tokenizer revision, input IDs, attention mask, and length limit. Decode the non-padding IDs. Check that the instruction survives in both cases and the full answer only in the early case. Compare with the untruncated IDs to catch any second truncation during batch preparation. The notebook uses whitespace tokens and a 20-token limit solely to expose this accounting. A trained tokenizer generally splits words differently.
The word-token fixture produces the following visibility results. These are checks of the supplied input, not observations of generated text.

Version	Instruction survives	Complete gate sentence survives
Early answer	Yes	Yes
Late answer	Yes	No
Shortened control	Yes	Yes

Keep the checkpoint, instruction, decoder settings, and output cap fixed. Allow enough output tokens for the requested sentence. Record an omitted gate and a wrong gate separately. A correct guess after evidence removal can match the reference without being supported by the supplied text.

If the late input lacks the fact, preprocessing removed access to that source evidence. If the early input contains both instruction and fact but the model omits the answer, the failure lies after preprocessing: encoding, conditional prediction, or decoding. These observations alone do not identify which model stage failed. Shorten only the irrelevant material in the late version, verify that the answer survives, and regenerate. Recovery supports the truncation explanation. If omission persists, report both visibility and model behavior. Repeat with several notices and gate values. The code above verifies visibility only; it makes no claim that a model was run or recovered.

12. Plan a comparison (optional)

Compare a compact pretrained encoder-decoder with the same checkpoint after fine-tuning on ferry notices and human summaries. Each task requests the cause of a delay and the actual departure time. A paired test then reveals whether adaptation restores a missing time while adding an unsupported cause.

Group by underlying event before splitting. All updates about one departure, duplicates, and lightly rewritten notices stay together. Use approximate 70/15/15 train/validation/test proportions by group and record IDs and counts. Train only on training pairs. Start runs from the same base weights and save the model and tokenizer revisions, prompts, seeds, update budget, and checkpoints. Hold test groups aside until all choices are fixed.

One concrete validation rule selects the checkpoint with highest mean per-notice ROUGE-L F1 under fixed decoding, with the earlier checkpoint winning a tie. Monitor source support alongside that score. Freeze the same source format, input truncation, beam width (for example four), length penalty, output cap (for example 64 generated tokens), and EOS rule for both test conditions. Changing each model's decoding independently would confound a claim about weight adaptation alone.

Use one declared tokenization and sentence-level ROUGE-L implementation, with F1 weighting and empty-input conventions stated. Average per-notice scores and also report lengths and empty outputs. Good paraphrases may score lower; fluent inventions can score higher. Source checking supplies a separate view.

Before reading outputs, list the required source facts per notice. Mark which are correctly conveyed. Independently divide each output into factual claims and label them supported, contradicted, or unverified against the source. Hide checkpoint identity from annotators, use the same rules, and resolve disputes against the notice. A correct time plus an unstated inspection cause conveys one required fact while adding an unverified claim.

Report required-fact coverage and unsupported-claim share separately. Add the respective numerators and denominators across notices before dividing, keeping contradicted and unverified counts visible. Also report the proportion of notices with any unsupported addition. An empty summary has zero coverage when facts are required, but no claim-level denominator; record it separately rather than calling it perfectly faithful. The illustrative annotation below has two of three required facts and one unverified claim among four claims. It gives coverage $2/3$ and unsupported share $1/4$. These are example counts, not measured model results.

Show paired differences on the same test notices, with examples of both improvements and regressions. For uncertainty, resample whole event groups and keep both checkpoints' outputs for every sampled notice. This preserves dependence among related notices and pairing across systems. The conclusions apply to the recorded data, annotation policy, and generation configuration.